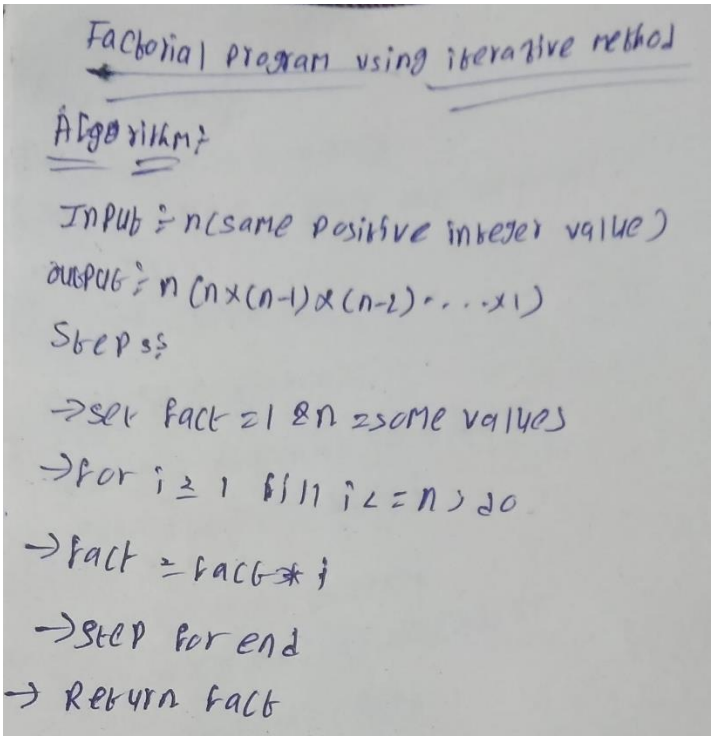
PRACTICAL-4

AIM: Implementation and Time analysis of factorial program using iterative and recursive method.

Factorial Using Iterative Method

ALGORITHM:



CODE:

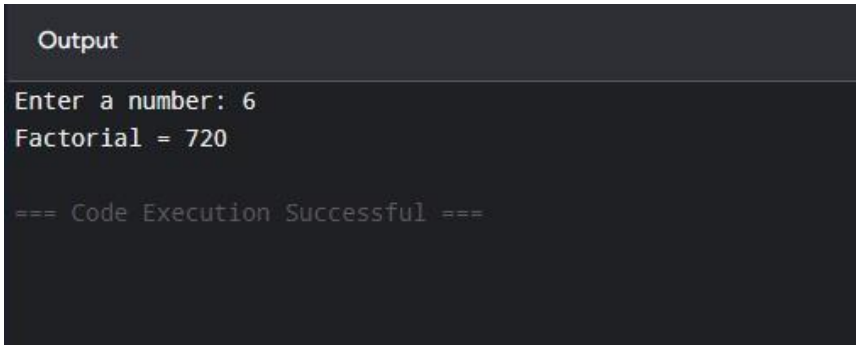
```
#include <iostream>
using namespace std; int
factorial(int n)
{
   int fact = 1;
   for(int i = 1; i <= n; i++)
   {       fact =
fact * i;
   }    return
fact; }
```

```
int main()
{    int n;
   cout << "Enter a number: ";
cin >> n;

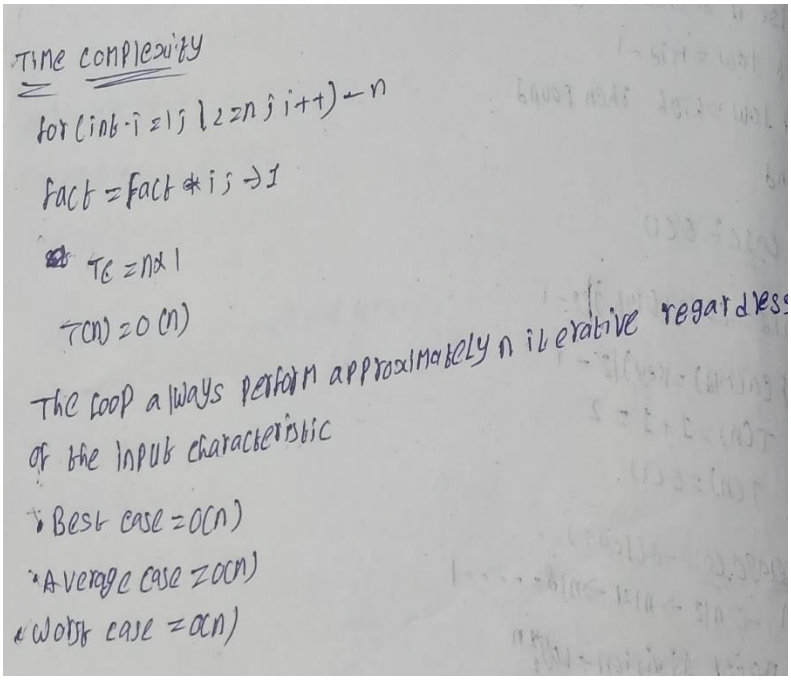
   cout << "Factorial = " << factorial(n);

   return 0;
}
```

OUTPUT:

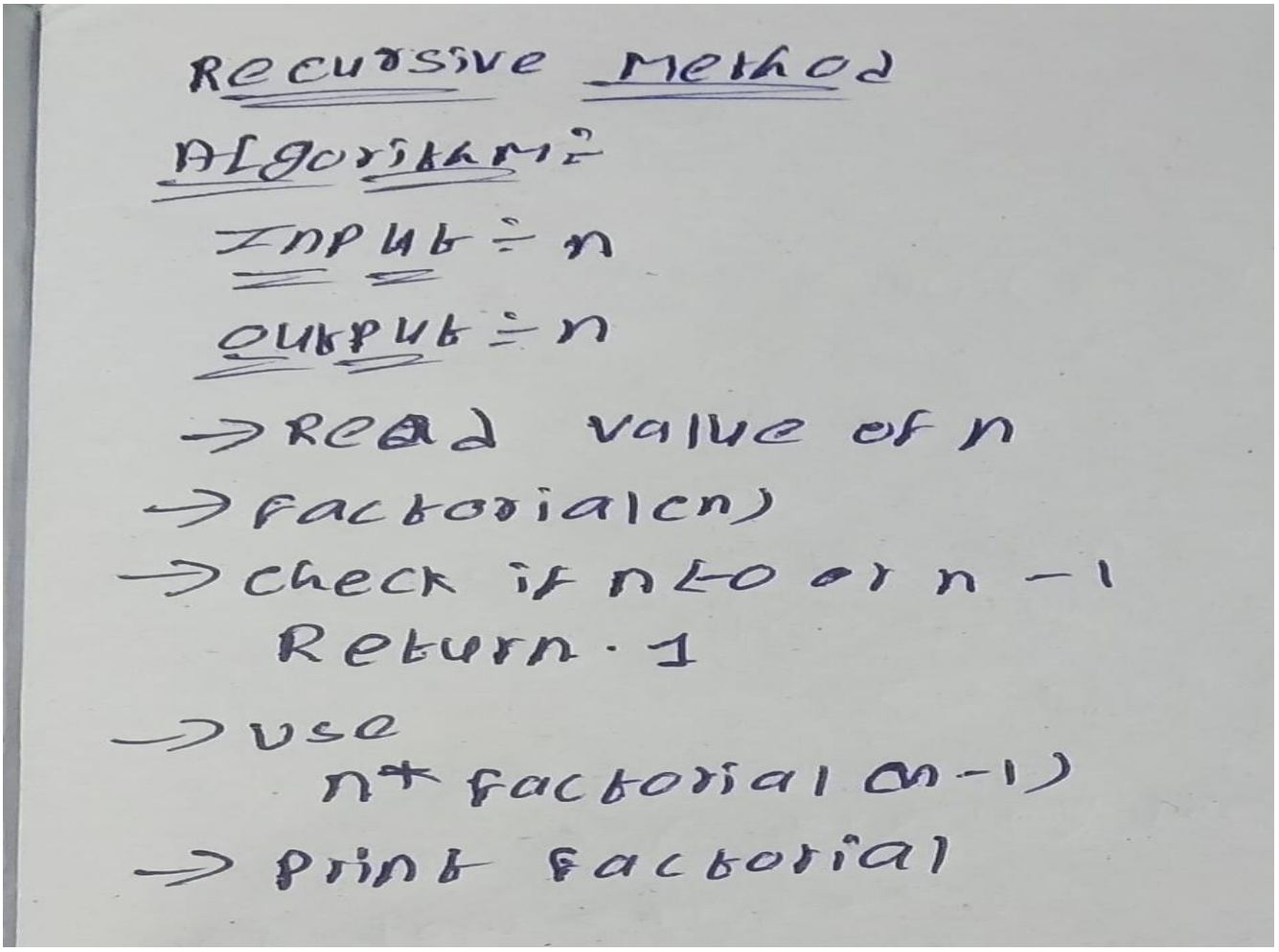


TIME COMPLEXITY:



Factorial Using Recursive Method:

Algorithm:



CODE:

```
#include <iostream>
using namespace std; int
factorial(int n)
{    if(n == 0 || n ==
1)
   {
      return 1;
   }
   return n * factorial(n - 1);
```

```
} int
main() {
int n;
   cout << "Enter a number: ";     cin >>
n;     cout << "Factorial = " <<
factorial(n);     return 0;
}
```

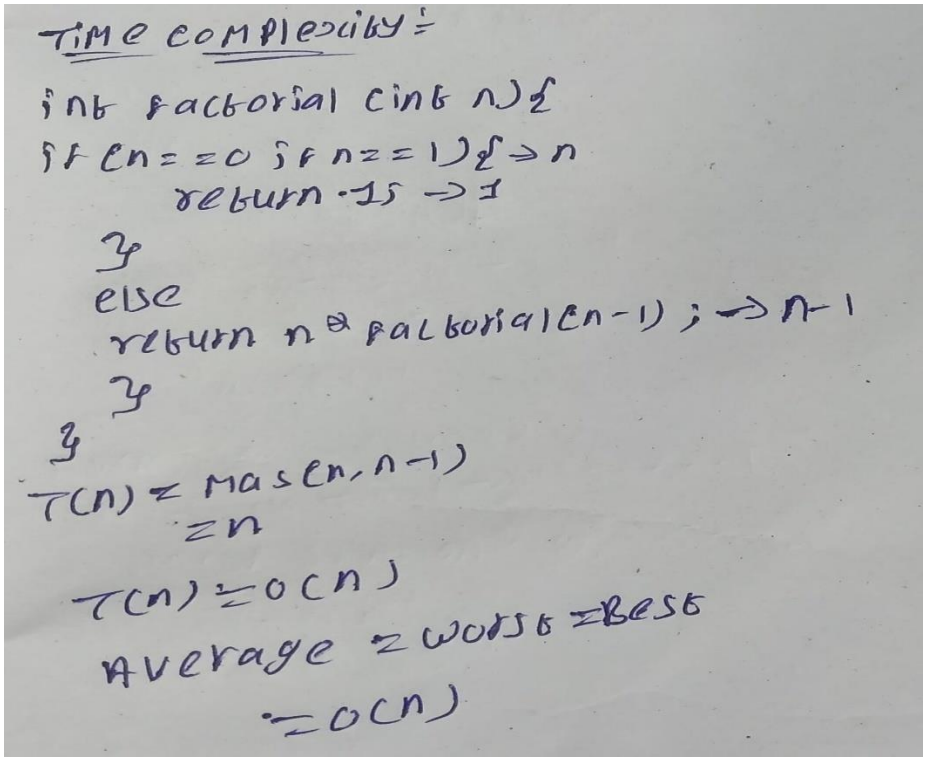
OUTPUT:

```
Output

Enter a number: 5
Factorial = 120

=== Code Execution Successful ===
```

TIME COMPLEXITY:



CONCLUSION:

The factorial program was successfully implemented using both iterative and recursive methods. Both methods produce the same result, but the recursive method uses function calls and stack space. The time complexity of both methods is O(n), making them efficient for calculating factorials.